

Evaluation and a New Benchmark for Defect Detection in Materials Fabrication

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Abstract

Defect detection in materials, especially in nuclear applications, remains a critical yet challenging endeavor due to the diversity and complexity of potential defects. While prior work has shown that standard CNN based methods can succeed when trained on large scale defect datasets, few-shot learning scenarios have not been thoroughly explored an issue that arises acutely when only limited, expert labeled examples are available. Building on existing research, we compare conventional semantic segmentation models with few-shot and multimodal approaches, leveraging two established benchmarks and introducing a new, open-source stainless steel defect dataset.

1. Introduction

Deep learning models have demonstrated significant utility in materials science, particularly for classifying defects. However, many approaches struggle to generalize in few-shot scenarios. In nuclear materials research in particular, the accurate detection of defects is critical due to the stringent safety and performance requirements of nuclear applications. In nuclear materials, computer vision techniques automate the characterization of microstructure and detect defects in fabricated materials. However, the diverse and often complex nature of these structures, surfaces, and defects often poses significant challenges for traditional vision models. The scarcity or complete absence of comprehensive datasets representing the full spectrum of real-world defects further exacerbates these challenges.

In this work, we expand on recent work in the field [5] to explore more complex learning models for these tasks. Specifically, our contributions include:

- Creation of **Stainless-25**, an open source dataset for materials fabrication analysis and defect detection, consisting of images and semantic masks - sample in Figure 1.

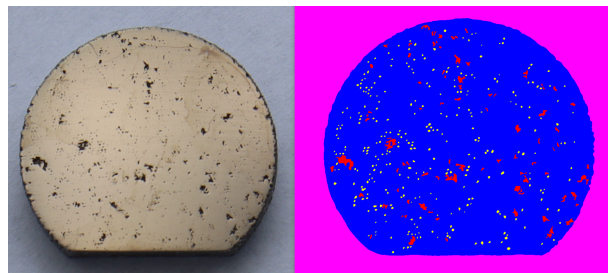


Figure 1. Sample data from Stainless-25 dataset: image of sliced sample (left) and semantic mask (right). Key: magenta=background, blue=stainless steel material, red=lack-of-fusion defect, yellow=keyhole defect

- Evaluation of semantic segmentation models, including few-shot prototypical networks and multimodal foundation models, with evaluation on established materials research benchmarks.

2. Related Work

Publicly available materials datasets are typically limited in both size and resolution, making them insufficient for many data-intensive machine learning tasks in materials science [9]. This constraint becomes especially problematic for defect identification and localization, which require highly granular annotations that distinguish subtle anomalies from normal variations [6]. As a result, researchers face significant barriers when training high-capacity models, since obtaining large, finely labeled datasets is often prohibitively expensive or impractical. MVTec-AD [2, 3], a widely used benchmark for industrial defect detection, addresses some of these challenges by providing high-resolution images with pixel-precise annotations, making it a valuable resource for evaluating anomaly detection methods. However, it is limited in material diversity and defect types, making it less generalizable to more complex real-world applications[1].

3. Materials Analysis & Defect Detection

Prior work [5] in this domain highlights the capacity for convolutional neural networks (CNNs) for materials analysis, and particularly leverage well-established models such as DeepLabV3[4]. In this work, comparison of this model is made with a few-shot architecture PRNet[10], a prototypical residual network. While this prototypical model addresses the data concerns encountered in the baseline model, it cannot enable the multimodality needed to incorporate the robust metadata of the materials fabrication process into the training process for the model. For this reason, we additionally include in our experiments the latest segment anything model (SAM2)[7], a foundation transformer model with this capability.

To evaluate the efficacy of these models on experimental materials data as well as their generalizability across materials and tasks, we leverage 3 datasets: two established benchmarks and one dataset gathered in the course of this research on materials fabrication.

3.1. Experimental Data Collection: Stainless-25

The proposed dataset consists of defects within additively manufactured 316L stainless steel nuclear fuel rod samples. These rods, fabricated through laser powder bed fusion (LPBF), were intentionally produced with varied process parameters in order to induce a range of defects, primarily lack of fusion (LOF) and keyhole defects.

To create the defect dataset, each fabricated rod was sectioned, yielding ten distinct samples. From these, a non-sequential selection process was implemented, choosing a maximum of two samples per rod. This selection strategy aimed to provide a diverse representation of potential defect occurrences while minimizing bias from localized manufacturing variations.

Image acquisition was performed using a mobile phone camera, resulting in images with a targeted resolution of 2000x2000 pixels. However, it is important to note that the raw dataset exhibits variability in image resolution, with observed resolutions ranging from approximately 900x900 pixels to 4K. This variation introduces a potential source of inconsistency within the dataset.

3.2. Segmentation

DeepLabV3 [4] is a widely used semantic segmentation model with atrous convolutions. By adjusting the dilation rate, it can effectively capture multi-scale contextual information without significantly increasing computational overhead. DeepLabV3 is particularly suited for large-scale datasets in which defects may appear at different resolutions or scales. In this work, its proven robustness and flexibility make it a natural baseline for comparing segmentation performance across challenging material defect images.

Class	# of images
background	29
material	29
lack of fusion defects	28
key hole defects	29
# labeled images	29
# unlabeled images	306
Total # images	336

Table 1. Dataset description of the Stainless-25 materials fabrication experiments, released with this publication. Data was gathered from experiments where different defects were intentionally induced in order to assess conditions & effects.

PRNet [10] is designed with a focus on few-shot learning, enabling accurate segmentation when only a limited number of labeled examples are available. Its architecture incorporates prototype representations, which distill the essential features of the target class from small annotated sets. This characteristic is particularly beneficial for materials science tasks where collecting defect labels is costly and time-consuming. By effectively leveraging sparse supervision, PRNet helps reduce the annotation burden and accelerates the model development cycle.

SAM2 [7] is an iteration of the “segment anything model”. Unlike the strictly image based approaches, SAM2 integrates multimodal inputs such as text prompts or other auxiliary data—into the segmentation process. This allows the model to incorporate high-level cues (e.g., defect descriptions or material properties) when delineating regions of interest. In scenarios where defects are subtle or visually ambiguous, the additional semantic context can significantly enhance segmentation accuracy. SAM2’s reliance on both visual and textual clues underscores a growing trend toward more holistic, context-aware segmentation strategies in materials science.

4. Experiments in Defect Detection

Prior to experimentation, images in each dataset were resized to a minimum of 512x512. An example set of results for the following experiments is displayed in Figure 2.

4.1. Experiments on Graphite-23

The Graphite-23 dataset [8, 9] is a benchmark for semantic segmentation of micrographs of graphite material.

The dataset consists of 12 images of sizes 1194x1194 to 3594x3594, each with pixel-level labels for 3 classes: void/pore, graphite material, and epoxy/resin. Since full pixel-level annotations are available, the dataset is used to evaluate each model’s performance in material texture seg-

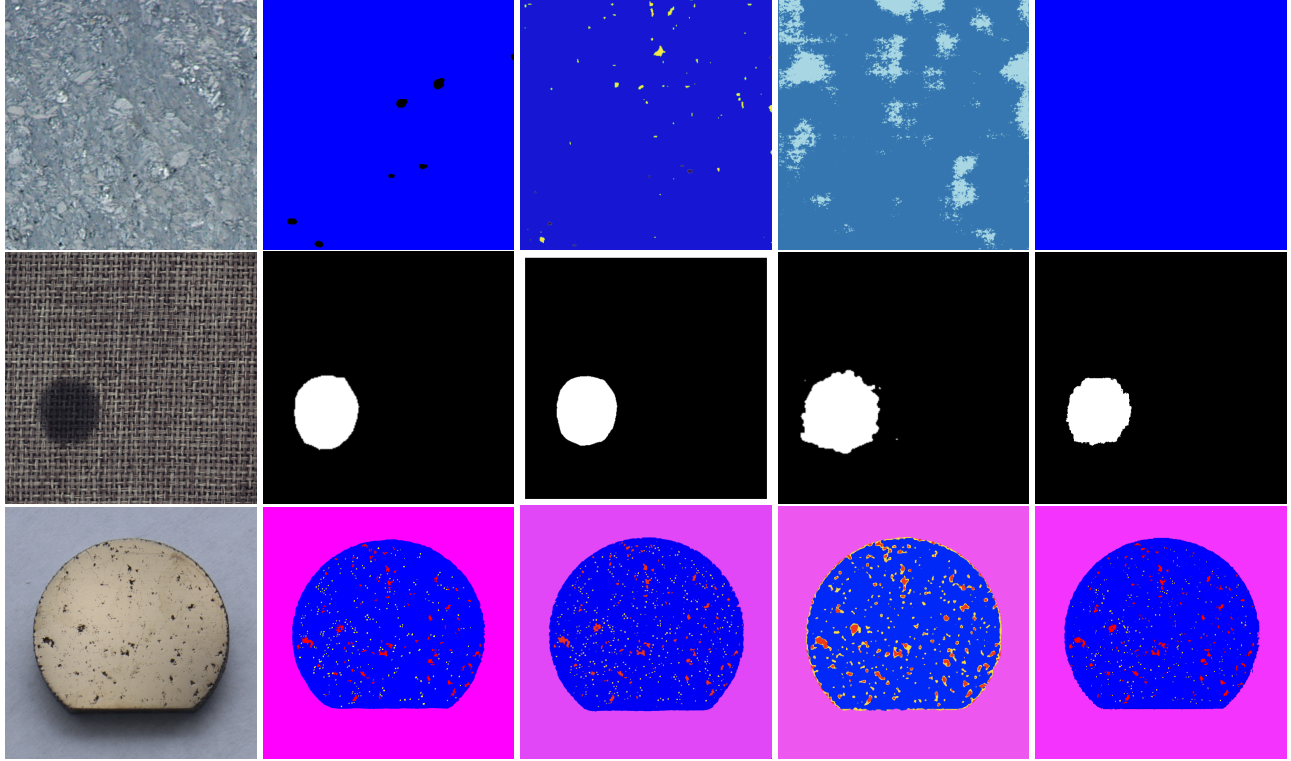


Figure 2. Results of each model on each dataset. From left: original image, ground truth, deeplabv3, PRNet, SAM2. Top row: Graphite-23[9]. Middle row: MVTec-AD[2, 3]. Bottom row: Stainless-25 (ours).

mentation. Table 2 compares the mean intersection over union (mIoU) scores of the three evaluated models.

Each training pipeline follows a shared configuration for key hyperparameters, including Adam-based optimizers, an initial learning rate of $1e-4$, and learning rate schedulers based on Cosine Annealing. With the exception of SAM2, both DeepLabV3+ and PRNet utilize a ResNet50 encoder and a batch size of 16.

For Graphite23, DeepLabV3+ achieved the best performance. As is often the case with CNN-based models, Adam facilitated faster convergence and produced less noisy results for both DeepLabV3+ and PRNet. Model performance decreased when AdamW was used. Conversely, SAM2 performed well with AdamW. However, even when AdamW was used with a weight decay of $1e-3$, DeepLabV3+ still slightly outperformed the other two models.

In an attempt to improve performance for PRNet and SAM2, a Cosine Annealing with Warm Restarts scheduler. This change appeared to benefit both models by taking advantage of periodic learning rate resets, yet their overall performance remained lower than expected. However, this discovery led us to adopt this scheduler for the remaining two datasets.

As indicated by our results, SAM2 was our second-best performer. Leveraging its multimodal capabilities, we used

Class	deeplabv3[4]	PRNet[10]	SAM2[7]
Graphite	0.8271	0.7320	0.8199
void/pore	0.7164	0.0804	0.5818
resin/epoxy	0.6883	0.3053	0.6444
Average overall	0.7439	0.3726	0.6820

Table 2. Evaluation (mIoU) on the Graphite-23[9] dataset of micrographs of graphite material.

50 points for training. It became apparent that increasing the number of points would often improve performance but at the cost of higher memory usage. It is worth noting that this dataset presents a different category of defect detection compared to Stainless-25 or MVTec, underscoring the importance of domain factors in model performance.

4.2. Experiments on MVTec-AD

The MVTec Anomaly Detection (AD) dataset [2, 3] is a large scale resource commonly used for unsupervised anomaly detection, comprising 15 texture categories each with five defect types and varying numbers of image mask pairs. For these experiments, only five texture categories (carpet, grid, leather, and tile) were selected due to their relevance for defect detection in materials. DeepLabV3+ and

Class	deeplabv3[4]	PRNet[10]	SAM2[7]
carpet texture	0.82	0.993	0.99
carpet defect	0.00	0.422	0.88
grid texture	0.92	0.992	0.99
grid defect	0.00	0.205	0.90
leather texture	0.83	0.997	0.99
leather defect	0.00	0.484	0.92
tile texture	0.82	0.979	0.99
tile defect	0.01	0.679	0.95
wood texture	0.39	0.990	0.99
wood defect	0.00	0.781	0.94
Average (defects)	0.00	0.514	0.918
Average overall	0.378	0.752	0.954

Table 3. Evaluation (mIoU) on the texture classes of the MVTecAD dataset[2, 3].

SAM2 were trained following a loop inspired by PRNet’s approach, concentrating on one material class and its five defect types at a time before proceeding to the next. SAM2 delivered the best performance overall, particularly when using 40 to 60 prompt points, and benefited from knowledge gained during previous training on other material classes. PRNet was the second-best performer, leveraging k-means clustering of unlabeled data in its fusion layers and multi-scale attention mechanisms; however, excessive multi-scale attention could degrade results. To stabilize training and improve gradient flow, LayerNorm was applied before the attention module. Although these architectural advantages gave PRNet an edge over DeepLabV3+, it still fell short of SAM2’s performance. Table 3 compares the mean intersection over union (mIoU) scores of the three models under evaluation.

4.3. Experiments on Stainless-25

In an effort to assess each model’s ability to generalize beyond synthetic scenarios, their performance was evaluated on the Stainless-25 dataset. Table 4 summarizes the key results. SAM2 demonstrated the strongest results, consistently outperforming the other approaches on the majority of metrics. This superior performance suggests that SAM2 is particularly well-suited for tasks involving stainless steel data, highlighting robust generalization capabilities in real-world experimental settings.

In these experiments, the baseline segmentation models were configured using a ResNet-50 backbone pretrained on ImageNet, an Adam optimizer with a learning rate of 1×10^{-3} , and a weight decay of 1×10^{-4} . A cosine annealing scheduler was employed to dynamically adjust the learning rate over 100 epochs, and data augmentation was enabled to promote robustness. The batch size was set to 3, accommodating memory constraints typical in segmentation tasks. DeepLabv3 was trained under similar hyperpa-

Class	deeplabv3[4]	PRNet[10]	SAM2[7]
background	0.9908	0.98	0.99
material	0.9666	0.93	0.98
LOF defect	0.2990	0.32	0.85
keyhole defect	0.1183	0.06	0.71
Average (defects)	0.2087	0.035	0.78
Average overall	0.5937	0.448	0.883

Table 4. Evaluation (mIoU) on the Stainless-25 dataset, which includes classes for material as well as two types of defects: lack-of-fusion (LOF) and keyhole.

rameter settings and strategies, ensuring that any observed differences in performance could be attributed primarily to model architectures rather than inconsistencies in the training process.

5. Conclusions

In this work, we build upon recent advances in deep learning for materials characterization and prediction, in order to accelerate the development of optimized fabrication processes. We further contribute assessment of multimodal models to this task, and introduce a novel dataset for defect detection with labeled masks. To this end, we evaluate three types of architectures on three datasets: Graphite-23 [9] micrographs, MVTecAD [2, 3] defect detection in textures, and Stainless-25, our proposed open source dataset of polished stainless steel surfaces for defect identification.

We find that while there is potential for the more often discussed CNN architectures on basic analysis tasks, there are unique challenges in identifying defects or anomalies in textures or additively manufactured materials. The few-shot and multimodal models demonstrate a significant improvement on this defect identification and localization task. Our contributed Stainless-25 dataset for defect localization and identification introduces a critical benchmark for the community, enabling rigorous and standardized evaluation of future machine learning methodologies.

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